

On-the-Fly Non-adiabatic Molecular Dynamics Reveals Dissociation Mechanisms of Multiply-Charged Molecules

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(Dated: March 2, 2026)

Understanding the dissociation of multiply charged molecules is crucial yet challenging due to complex multi-body correlations and non-adiabatic dynamics. Conventional *ab initio* molecular dynamics simulations commonly struggle to capture excited electronic states and intricate electron-nuclear coupling, particularly in distinguishing sequential and non-sequential dissociation pathways. To address these limitations, we present a combined theoretical and experimental study on the dissociation mechanisms of CO_2^{3+} . Our approach employs on-the-fly non-adiabatic molecular dynamics simulations, utilizing Dalitz plots and Newton diagrams to differentiate dissociation channels. The simulations reproduce the key dynamical signatures observed via a COLTRIMS reaction microscope, enabling the state-resolved disentanglement of overlapping spectral features. This work provides mechanistic insights into the fragmentation dynamics, refining the interpretation of experimental data and offering an in-depth understanding of non-adiabatic molecular dynamics.

Coulomb Explosion Imaging (CEI) [1–19], first shown in 1986 [1], has revolutionized the field of ultrafast molecular dynamics. It offers unparalleled insights into the geometric and temporal evolution of molecular systems on femtosecond-attosecond timescales. CEI works via controlled multiple ionization, triggering Coulomb-driven dissociation that retains the molecular spatial correlations at fragmentation. By accurately measuring the three-dimensional momentum distributions of ionic fragments, it allows for the reconstruction of the initial molecular geometry and subsequent dissociation pathways, which makes CEI crucial for understanding complex molecular dynamics [10, 18].

CEI of multiply charged molecule has been experimentally investigated through various excitation mechanisms, including intense laser fields [3–8, 20], X-ray irradiation [9–12], ion [13–18, 21] and electron collisions [19]. The three-dimensional momentum vectors of resulting ionic fragments are precisely measured in coincidence using Cold Target Recoil Ion Momentum Spectroscopy (COLTRIMS) [22]. The fragment momentum distributions encode a complete set of information pertaining to the excitation process, including the population of specific electronic states and the intricate non-adiabatic coupling between electronic and nuclear degrees of freedom. These multidimensional correlations are ultimately reflected in the asymptotic momentum distributions of the ionic fragments. Through sophisticated reconstruction algorithms, these momentum distributions enable the determination of fundamental dynamical parameters [6, 10, 18, 19], including the initial molecular geometry prior to fragmentation and the temporal evolution of dissociation pathways.

As a prototype system of three-body fragmentation, the process of CO_2^{3+} has been extensively studied with

synchrotron radiation [23], electron impact [19, 24, 25], highly charged ion beams [15, 21, 26], as well as intense laser fields [6, 20]. The mechanism of nonsequential and sequential dissociation processes were clearly revealed experimentally. To provide a deeper microscopic interpretation of these experimental phenomena, theoretical frameworks have concurrently undergone transformative developments. Early theoretical efforts leveraged static point charge models [27] and precomputed potential energy surfaces (PESs) [6, 19] to extract kinetic energy release (KER) distributions and infer structural parameters at the moment of fragmentation. A significant paradigm shift emerged with the introduction of on-the-fly *ab initio* molecular dynamics [28], which circumvents the need for precomputed PESs. This approach, utilizing the Atomic Center Density Matrix Propagation (ADMP) method [29–31], enables direct simulation of multiply charged molecular dissociation from their ground states, offering unprecedented dynamical resolution without a priori assumptions about the PES.

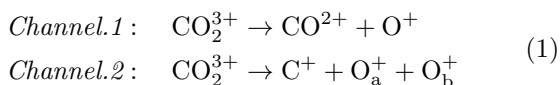
Despite these substantial advancements in the theoretical understanding of the dissociation dynamics of CO_2^{3+} , several critical limitations remain unresolved. For instance, on-the-fly molecular dynamics methods rooted in the Born-Oppenheimer (BO) approximation suffer from an intrinsic limitation: their dependence on a single electronic state inherently restricts their ability to model non-adiabatic transitions and multi-state coupling phenomena, rendering them unsuitable for systems dominated by strong non-adiabatic interactions. Although PES-based dynamics theoretically accommodate multi-state processes, their practical implementation requires precomputed high-dimensional PESs—a computationally prohibitive task for complex molecular systems.

In this Letter, we address these challenges by inves-

tigating the dissociation of CO_2^{3+} using an on-the-fly non-adiabatic dynamics framework. This approach allows for the direct treatment of multi-state couplings and excited-state evolution without precomputed potential energy surfaces. The simulations were conducted using the SHARC (Surface Hopping including ARbitrary Couplings) approach [32, 33] interfaced with OpenMolcas [34] for electronic structure calculations at the CASSCF level [35, 36]. By combining these trajectory-based simulations with Coulomb-explosion observables, we distinguish concerted and sequential pathways and relate state preparation to specific KER and Dalitz/Newton signatures. While the essential simulation parameters are summarized below, the detailed theoretical framework is provided in SUPPLEMENTAL MATERIAL NOTE 1 [37].

In the molecular dynamics simulations, the initial electronic states were selected within the adiabatic representation. This set corresponds to the five lowest doublet states of CO_2^{3+} (D_0 – D_4), where D_0 denotes the ground state and D_1 – D_4 represent the four lowest excited states. The simulations were initiated by thermally sampling the neutral CO_2 molecule (comprising one carbon and two oxygen atoms) at 300 K. A total of 3500 initial conditions, encompassing nuclear positions and velocities (see SM NOTE 1 for more details), were generated by statistically sampling the vibrational and rotational populations according to the Boltzmann distribution. These trajectories were then propagated for a total simulation time of 500 fs with a fixed time step of 0.5 fs.

After 500 fs of propagation, each trajectory was analyzed to identify the final distribution of fragments. Using a bond distance criterion of $R > 2.5 \text{ \AA}$ at 500 fs to define dissociation, two dissociation channels were categorized as:



O_i^+ ($i=a, b$) indicates the oxygen ion fragment.

To minimize *a priori* assumptions, the initial sampling was governed by a single criterion: ensuring statistical stability of the spectral features for every electronic state. Specifically, over 500 trajectories were propagated for each of the D_0 – D_4 states to obtain robust, intrinsic KER profiles. These profiles subsequently served as fixed basis functions to fit the experimental spectrum, allowing the relative populations to be derived *post hoc*. In this protocol, the trajectory-level non-adiabatic dynamics are obtained fully *ab initio*; only the state populations are adjusted *a posteriori* to reflect the specific experimental preparation and detection. Fig. 1 systematically illustrates the proportions of two dissociation pathways of CO_2^{3+} , *Channel.1* ($\text{CO}^{2+} + \text{O}^+$) and *Channel.2* ($\text{C}^+ + \text{O}_a^+ + \text{O}_b^+$), using a donut chart and a horizontal bar chart. The donut chart reveals the overall fragment

proportions, with *Channel.1* accounting for 20.4% and *Channel.2* comprising 79.6%. The horizontal bar chart further deconstructs the contributions of different electronic states to fragment formation, where the x -axis represents the number of dissociation trajectories, and colors correspond to initial electronic states. For the ground state (D_0), about 23% of the total trajectories undergo three-body fragmentation, which is consistent with the 27% yield reported in Yang’s simulation [28] using the ADMP approach [29–31].

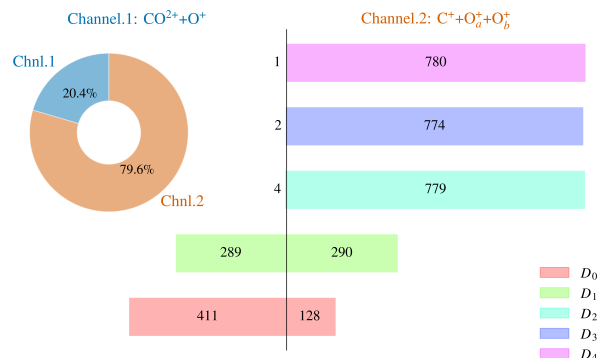


FIG. 1. Proportions of the two dissociation pathways $\text{CO}^{2+} + \text{O}^+$ (*Channel.1*) and $\text{C}^+ + \text{O}_a^+ + \text{O}_b^+$ (*Channel.2*). The horizontal bar chart shows the distribution of fragments initially located in different electronic states (D_0 to D_4), with colors representing the respective excitation Doublet states.

The trajectories of *Channel.1* predominantly arise from the two lowest electronic states (D_0 , D_1), while the three-body dissociation (*Channel.2*) is mainly formed from highly-excited states (D_2 , D_3 , D_4). These results highlight the significant role of excitation energy in determining dissociation pathways: lower electronic states favor two-body fragments, while highly-excited states are more conducive to three-body dissociation. To validate the experimentally observed three-body dissociation mechanisms and gain deeper insights into intermediate processes, we present Dalitz plots [15, 38] and Newton diagrams [15, 19, 28] for CO_2^{3+} (Mathematical details are provided in SM NOTE 2.), as shown in Fig. 2. Figs. 2(a)–(d) display the Dalitz plots at 500 fs: (a) shows the experimental results, obtained using a COLTRIMS reaction microscope combined with a pulsed photoemission electron source (see SM NOTE 5 [37]), while (b)–(d) present the simulated distributions, including all trajectories (b), contributions from low-lying states (D_0 , D_1) (c), and from highly excited states (D_2 – D_4) (d). Each point in the plots represents a specific momentum correlation among the three fragments at the moment of breakup [15, 19, 28]. For instance, the bottom-center point $(0, -1/3)$ corresponds to linear fragmentation, where the two O^+ ions recoil in opposite directions and the C^+ ion remains at rest. Points near $x = 0$ indicate bending-type dissociation linked to molecular vibra-

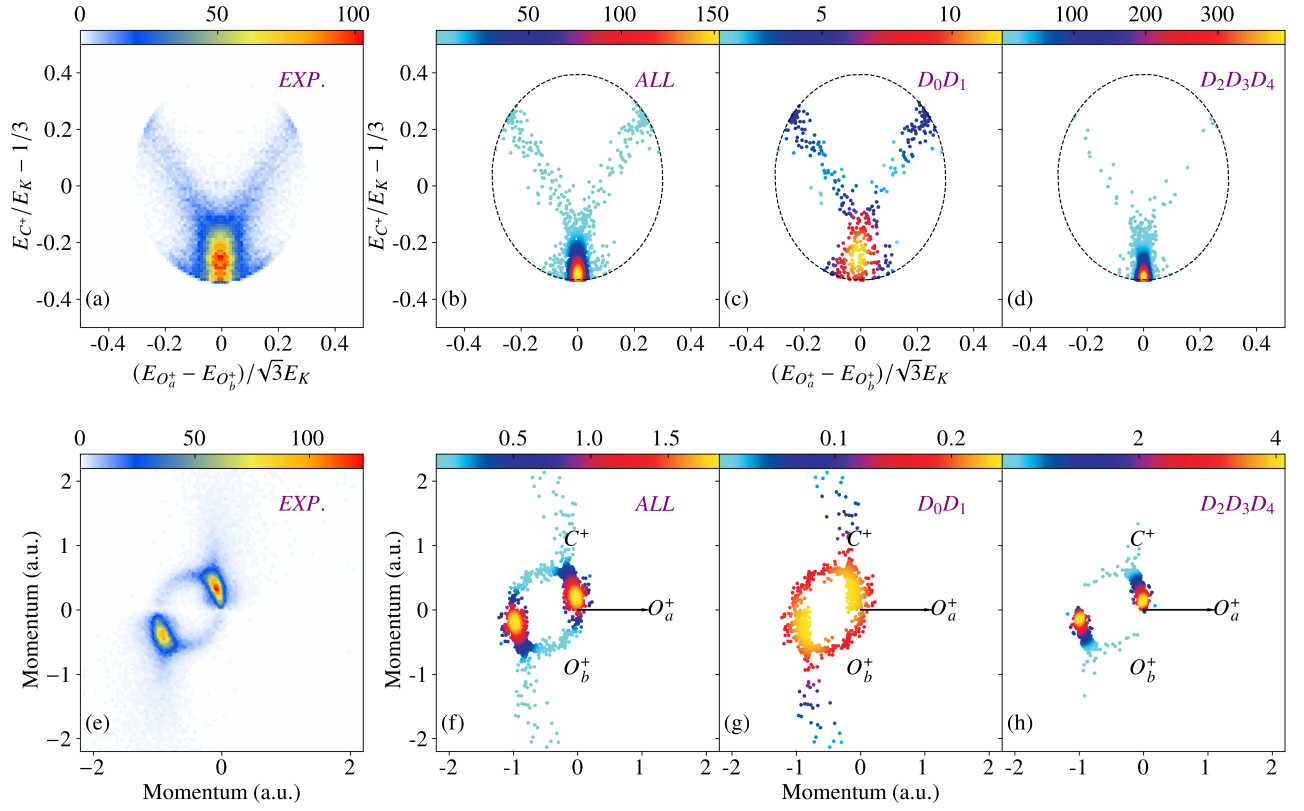


FIG. 2. Combined Dalitz plots and Newton diagrams for the dissociation dynamics of CO_2^{3+} . The color scale represents the density distribution, where each point corresponds to a single trajectory or experimental event. Panels (a) and (e) depict the experimental results (EXP.) of the Dalitz plot and the corresponding Newton diagram, respectively, demonstrating the actual observed dissociation patterns of CO_2^{3+} . Panels (b–d) show simulated Dalitz plots, with (b) representing the combined theoretical distribution (ALL states), (c) the contribution from the (D_0, D_1) states, and (d) from the (D_2, D_3, D_4) states. Panels (f–h) present the simulated Newton diagrams corresponding to the theoretical Dalitz plots above. These comparative plots highlight the agreement between theoretical predictions and experimental measurements, and they emphasize the significant influence of initial electronic states on the dissociation dynamics of CO_2^{3+} .

tional modes, while sequential fragmentation is reflected by the characteristic X-shaped structure [15, 19, 28]. The specific dissociation geometries associated with these regions are detailed in SM Figure S2, following the visualization style of recent studies [39–41]. The initial electronic state clearly influences the dissociation mechanism. In Fig. 2(c), low-lying electronic states (D_0, D_1) exhibit an X-shaped pattern, indicating a sequential dissociation pathway via an intermediate CO^{2+} species. In contrast, Fig. 2(d) reveals that highly excited states (D_2 – D_4) favor direct three-body dissociation, evident from the more symmetric and concentrated distribution.

Since on-the-fly MD explicitly resolves the temporal evolution of the fragmentation, it necessitates a rigorous kinematic definition to distinguish between concerted and sequential pathways. For the symmetric CO_2 system, this dynamical distinction is encoded in the energy partitioning between the two identical oxygen atoms. To quantify this, we introduce the energy asymmetry parameter A , which is kinematically equivalent to the

horizontal projection of the Dalitz plot. Defined as $A = (E_{O_a} - E_{O_b}) / (E_{O_a} + E_{O_b})$, this parameter serves as a robust metric: events with $|A| < 0.25$ are classified as concerted-like (symmetric energy sharing), while larger asymmetries signify sequential decay. Based on this criterion, the lower electronic states (D_0, D_1) exhibit coexisting pathways dominated by sequential fragmentation (concerted fraction $\sim 20\%$), whereas the higher states (D_2 – D_4) are predominantly concerted ($\gtrsim 95\%$). The corresponding distributions and discussions are detailed in Fig. S3 and SM Note 4.

Further insight into the dissociation dynamics can be gained from the Newton diagrams shown in Figs. 2(e)–(h), which correspond to the same datasets as the upper-row Dalitz plots. Panel (e) displays the experimental Newton diagram, while panels (f)–(h) show the simulations for all states, D_0D_1 , and $D_2D_3D_4$, respectively. In each Newton diagram, one O^+ ion is fixed along the x -axis with unit-normalized momentum; the momentum vectors of the C^+ and the other O^+ are plotted in the

upper and lower hemispheres, respectively. A ring-like structure in the Newton diagram provides direct evidence of the sequential fragmentation mechanism. This occurs via a two-step process: first, CO_2^{3+} dissociates into $\text{O}^+ + \text{CO}^{2+}$; then, after a time delay, the intermediate CO^{2+} further fragments into $\text{C}^+ + \text{O}^+$. Rotation of the intermediate group between steps leads to the ring formation. The jet-like structure arises due to the indistinguishable labeling of the two O^+ ions. In Fig. 2(h), the absence of a ring indicates that for high-lying states ($\text{D}_2\text{--D}_4$), the fragmentation occurs via a concerted Coulomb explosion, i.e., a direct three-body breakup.

Fig. 3 depicts the simulated time evolution of the bond lengths $R_{\text{C--O}_a}$ (first row), $R_{\text{C--O}_b}$ (second row), and the bond angle $\theta_{\text{O}_a\text{--C--O}_b}$ (third row) of CO_2^{3+} . The first column presents 3500 trajectories corresponding to all five electronic states. The second and third columns represent the trajectories for the two low-lying electronic states (D_0 , D_1) and the three highly-excited electronic states (D_2 , D_3 , D_4), respectively. Regarding the bond length $R_{\text{C--O}_a}$ in the first column of Fig. 3, the majority of the trajectories dissociate to a length of 10 Å within 130 fs. However, a significant portion of the remaining trajectories oscillate around the equilibrium position and do not undergo dissociation. Only a small fraction of the trajectories persist beyond 130 fs and eventually dissociate. The case is analogous for the bond length $R_{\text{C--O}_b}$. Concerning the time evolution of the bond angle $\theta_{\text{O}_a\text{--C--O}_b}$, it is evident that the majority of the trajectories are distributed within the range of $100^\circ - 180^\circ$. Conversely, a small subset of the trajectories exhibits a turning-point behavior upon reaching 0° .

The combination of Fig. 2 (Dalitz plots and Newton diagrams) and Fig. 3 (trajectory evolution plots) enables the direct discernment of the mechanisms governing the dissociation process of CO_2^{3+} in its entirety. For the low-lying electronic states (D_0 , D_1), two distinct dissociation mechanisms coexist: two-body and sequential three-body dissociations. In the case of two-body dissociation, the stable products $\text{CO}^{2+} + \text{O}^+$ are generated. The sequential three-body dissociation, on the other hand, involves an intermediate species, CO^{2+} , which subsequently dissociates into $\text{C}^+ + \text{O}^+$. In the second column of Fig. 3, the time evolution of bond lengths $R_{\text{C--O}_a}$ and $R_{\text{C--O}_b}$ indicate that there is a probability of dissociation when the length of intermediate CO^{2+} reaches its maximum value during the oscillation. The turning-back behavior of $\theta_{\text{O}_a\text{--C--O}_b}$ directly reflects the rotational behavior of the intermediate CO^{2+} . The dynamic evolution will eventually reflect in the Dalitz plots and Newton diagrams for the three-body dissociation, as shown in second column of Fig. 2.

In contrast, for highly-excited states (D_2 , D_3 , D_4), direct three-body dissociation is the predominant process. This dissociation mechanism is marked by rapid fragmentation occurring within an extremely short time frame,

as clearly shown in the third column of Fig. 3. The divergence observed within the initial 50 fs is attributed to the different excited states, each of which is associated with a distinct PES. The Dalitz plot presented in Fig. 2(d) reveals an almost symmetric energy distribution among the fragments. This indicates a swift and uniform redistribution of energy during the dissociation process. The Newton diagram in Fig. 2(h) showcases a concentrated momentum separation. The trajectory evolution plots in the third column of Fig. 3 offer additional evidence to support this mechanism. The bond lengths $R_{\text{C--O}_a}$ and $R_{\text{C--O}_b}$ increase rapidly within 50 fs. Concurrently, the bond angle $\theta_{\text{O}_a\text{--C--O}_b}$ exhibits a strong density concentration in the large-angle region ($160^\circ\text{--}180^\circ$).

The KER spectra were analyzed using kernel density estimation (KDE). The complete data processing pipeline—covering binning, bandwidth selection, and normalization—is detailed in SM Note 3 [37]. Fig. 4 presents the results: the lower panels display the raw per-state histograms directly from the simulations, while the upper panels show the corresponding peak-normalized KDE curves ($I_{\text{max}} = 1$). For the three-body channel, the total theoretical KER (black curve) is a weighted sum of the $\text{D}_0\text{--D}_4$ contributions, based on the fitted populations.

In the two-body channel (left), the experimental KER is compared against individual per-state KDEs. The simulation reproduces the main experimental features, with the dominant contributions from D_0 and D_1 aligning with the observed peak positions.

In the three-body channel (right), the experimental data are compared with the state-weighted total KDE. The spectrum exhibits a broad intensity maximum in the 20–23 eV region rather than a well-resolved sharp peak. The weighted theoretical curve captures the overall lineshape and the main spectral weight in this energy range, which is predominantly contributed by the lower-lying states (D_0 and D_1). On the higher-energy side, the experimental distribution shows an extended tail or shoulder-like broadening. The state-resolved decomposition indicates that the spectral weight in the $\sim 24\text{--}26$ eV region mainly originates from the excited states ($\text{D}_2\text{--D}_4$), whereas the simulated spectrum decays more rapidly beyond this range. These systematic differences in linewidth and high-energy spectral weight suggest that additional physical contributions are present in the experiment beyond those included in the current model.

This discrepancy primarily arises from the restriction of the present simulations to the doublet spin manifold. In particular, the experimental spectrum at higher excitation energies likely involves energetically accessible higher spin multiplicities (quartets, sextets, etc.) and inter-multiplet couplings that are not explicitly accounted for here. A complete reproduction of the full experimental spectrum would therefore require incorporating these high-lying mixed-multiplicity configurations, which lies beyond the scope of the present

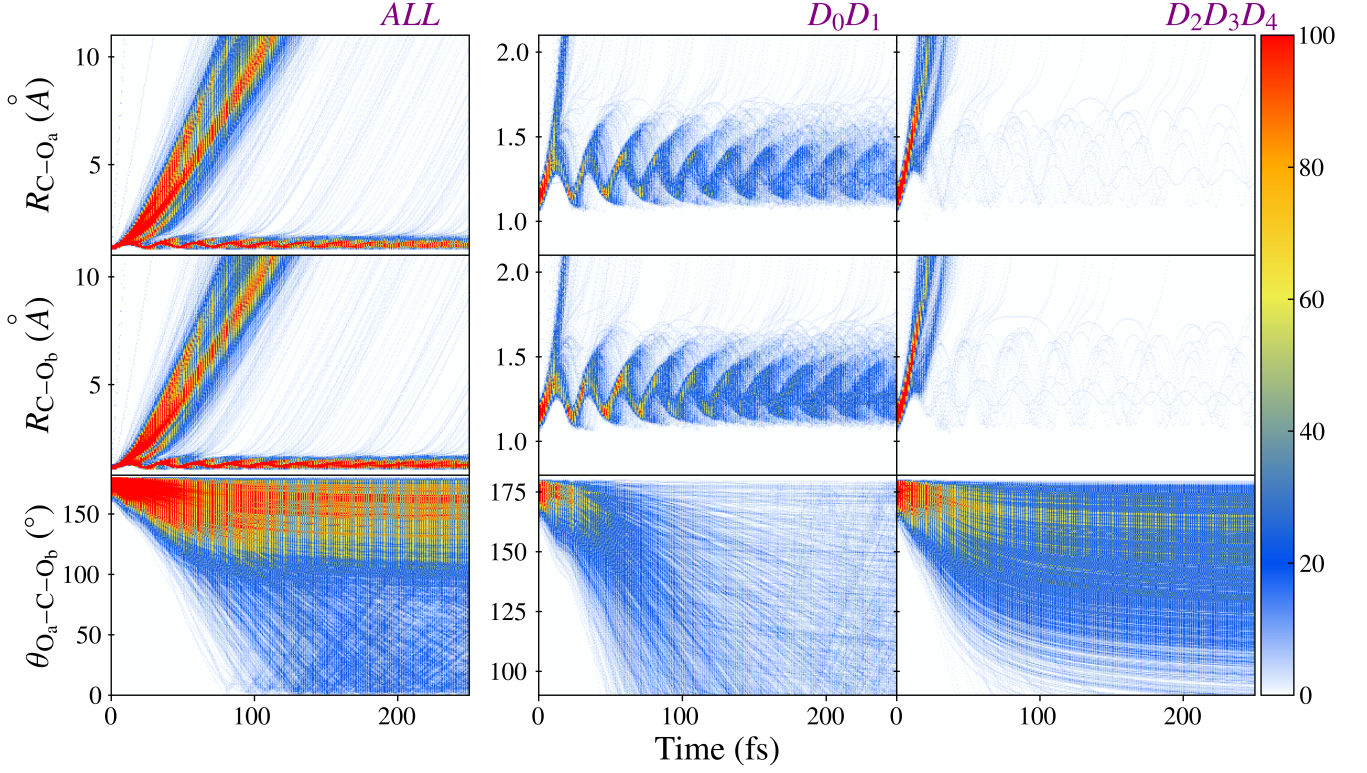


FIG. 3. Time evolution of bond lengths (R_{C-O_a} , R_{C-O_b}) and bond angle ($\theta_{O_a-C-O_b}$) for CO_2^+ dissociation (0–250 fs). Each column represents different trajectory groups: “ALL” (left column) includes all trajectories, (D_0, D_1) (middle column) corresponds to low electronic states, and (D_2, D_3, D_4) (right column) corresponds to highly-excited states.

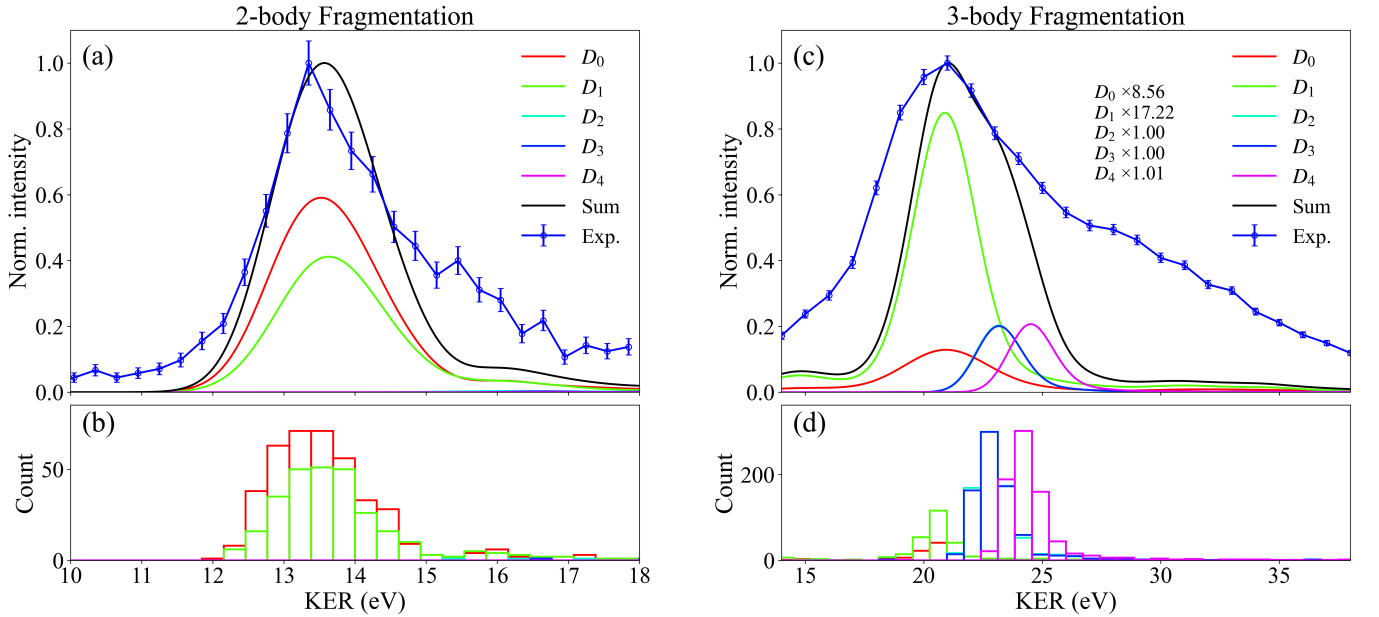


FIG. 4. Top: Peak-normalized KDE curves constructed from the corresponding lower-panel histograms on the same energy grid (Gaussian kernel, bandwidth = 0.35 eV); intensities are normalized to the maximum value ($I_{\max} = 1$). Bottom: Per-state histograms binned from simulated event-wise fragment kinetic energies. (a) Two-body KER: per-state KDEs overlaid with the experimental KER (no state weighting applied). (b) Two-body per-state histograms. (c) Three-body KER: state-weighted KDE (black) from D_0 – D_4 compared with the experimental KER. (d) Three-body per-state histograms.

work. Nevertheless, the current framework successfully reproduces the dominant spectral features and provides a clear, mechanism-level interpretation through state-resolved decomposition.

In summary, by employing on-the-fly non-adiabatic dynamics, this study reconstructs the experimental Dalitz plots, Newton diagrams, and KER spectra for the dissociation of CO_2^{3+} . By incorporating excited-state dynamics, the simulations reveal distinct dissociation pathways (two-body, sequential three-body, and direct three-body) governed by the initial electronic states. The methodological approach developed here is broadly applicable to similar dissociation systems and offers a robust platform for interpreting CEI experiments and guiding future theoretical and experimental investigations. This methodology not only bridges critical gaps in theoretical modeling but also establishes a unified framework for probing ultrafast dissociation mechanisms, with potential for advancing molecular fragmentation field.

Acknowledgments—The authors acknowledge financial support from the National Natural Science Foundation of China (Grants No. 12274337, No. 12374238, No. 12325406, No. 12574307, No. 12574280, No. W2512072, No. 12404305, and No. 125B2086), the Natural Science Basic Research Program of Shaanxi Province (No. 2025JC-JCQN-066), and the Fundamental Research Funds for the Central Universities (Grant No. GK202304006). The authors are grateful to Prof. Jianmin Yuan (Institute of Atomic and Molecular Physics, Jilin University) for valuable discussions.

Data Availability—The data that support the findings of this study are openly available in Zenodo at [42].

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- [1] Z. Vager, E. P. Kanter, G. Both, *et al.*, Direct determination of the stereochemical structure of CH_4^+ , *Phys. Rev. Lett.* **57**, 2793 (1986).
- [2] Z. Vager, R. Naaman, and E. P. Kanter, Coulomb explosion imaging of small molecules, *Science* **244**, 426 (1989).
- [3] F. Légaré, K. F. Lee, I. V. Litvinyuk, *et al.*, Laser coulomb-explosion imaging of small molecules, *Phys. Rev. A* **71**, 013415 (2005).
- [4] I. Bocharova, R. Karimi, E. F. Penka, *et al.*, Charge resonance enhanced ionization of CO_2 probed by laser coulomb explosion imaging, *Phys. Rev. Lett.* **107**, 063201 (2011).
- [5] M. Pitzer, M. Kunitski, A. S. Johnson, *et al.*, Direct determination of absolute molecular stereochemistry in gas phase by coulomb explosion imaging, *Science* **341**, 1096 (2013).
- [6] C. Wu, C. Wu, D. Song, *et al.*, Nonsequential and sequential fragmentation of CO_2^{3+} in intense laser fields, *Phys. Rev. Lett.* **110**, 103601 (2013).
- [7] M. Pitzer, G. Kastirke, M. Kunitski, *et al.*, Absolute configuration from different multifragmentation pathways in light-induced coulomb explosion imaging, *Chem. Phys. Chem.* **17**, 2465 (2016).
- [8] S. Bhattacharyya, K. Borne, F. Ziaee, *et al.*, Strong-field-induced coulomb explosion imaging of tribromomethane, *J. Phys. Chem. Lett.* **13**, 5845 (2022).
- [9] X. Li, A. Rudenko, M. S. Schöffler, *et al.*, Coulomb explosion imaging of small polyatomic molecules with ultrashort x-ray pulses, *Phys. Rev. Res.* **4**, 013029 (2022).
- [10] R. Boll, J. M. Schäfer, B. Richard, *et al.*, X-ray multiphoton-induced coulomb explosion images complex single molecules, *Nat. Phys.* **18**, 423 (2022).
- [11] E. Kukkk, K. Motomura, H. Fukuzawa, *et al.*, Molecular dynamics of xfel-induced photo-dissociation, revealed by ion-ion coincidence measurements, *Appl. Sci.* **7**, 531 (2017).
- [12] C. Cheng, L. J. Frasinski, G. Moğol, *et al.*, Multiparticle cumulant mapping for coulomb explosion imaging, *Phys. Rev. Lett.* **130**, 093001 (2023).
- [13] U. Werner, K. Beckord, J. Becker, *et al.*, 3d imaging of the collision-induced coulomb fragmentation of water molecules, *Phys. Rev. Lett.* **74**, 1962 (1995).
- [14] J. H. Sanderson, T. Nishide, H. Shiromaru, *et al.*, Near-coulombic behavior in the dissociative ionization of CO_2 due to impact by Ar^{8+} , *Phys. Rev. A* **59**, 4817 (1999).
- [15] N. Neumann, D. Hant, L. P. H. Schmidt, *et al.*, Fragmentation dynamics of CO_2^{3+} investigated by multiple electron capture in collisions with slow highly charged ions, *Phys. Rev. Lett.* **104**, 103201 (2010).
- [16] A. Khan, L. C. Tribedi, and D. Misra, Observation of a sequential process in charge-asymmetric dissociation of CO_2^{q+} ($q=4,5$) upon the impact of highly charged ions, *Phys. Rev. A* **92**, 030701 (2015).
- [17] A. Méry, V. Kumar, X. Fléchar, *et al.*, Coulomb-explosion imaging of carbon monoxide dimers, *Phys. Rev. A* **103**, 042813 (2021).
- [18] H. Yuan, Y. Gao, B. Yang, *et al.*, Coulomb explosion imaging of complex molecules using highly charged ions, *Phys. Rev. Lett.* **133**, 193002 (2024).
- [19] E. Wang, X. Shan, Z. Shen, *et al.*, Pathways for non-sequential and sequential fragmentation of CO_2^{3+} investigated by electron collision, *Phys. Rev. A* **91**, 052711 (2015).
- [20] C. Wu, C. Wu, Y. Fan, *et al.*, Three-body fragmentation of CO_2 driven by intense laser pulses, *J. Chem. Phys.* **142**, 124303 (2015).
- [21] S. Yan, X. L. Zhu, P. Zhang, *et al.*, Observation of two sequential pathways of CO_2^{3+} dissociation by heavy-ion impact, *Phys. Rev. A* **94**, 032708 (2016).
- [22] R. Dörner, V. Mergel, O. Jagutzki, *et al.*, Cold target recoil ion momentum spectroscopy: a ‘momentum microscope’ to view atomic collision dynamics, *Phys. Rep.* **330**, 95 (2000).
- [23] R. K. Singh, G. S. Lodha, V. Sharma, *et al.*, Triply charged carbon dioxide molecular ion: Formation and fragmentation, *Phys. Rev. A* **74**, 022708 (2006).
- [24] P. Bhatt, R. Singh, N. Yadav, *et al.*, Formation, structure, and dissociation dynamics of CO_2^{q+} ($q \leq 3$) ions due to impact of 12-keV electrons, *Phys. Rev. A* **85**, 042707 (2012).

- [25] X. Wang, Y. Zhang, D. Lu, *et al.*, Fragmentation of CO_2^{2+} in collisions with low-energy electrons, *Phys. Rev. A* **90**, 062705 (2014).
- [26] M. R. Jana, P. N. Ghosh, Bapat, *et al.*, Ion-induced triple fragmentation of CO_2^{3+} into $\text{C}^+ + \text{O}^+ + \text{O}^+$, *Phys. Rev. A* **84**, 062715 (2011).
- [27] C. Wu, G. Zhang, C. Wu, *et al.*, Dissociative double ionization of CO_2 induced by intense femtosecond laser pulses, *Phys. Rev. A* **85**, 063407 (2012).
- [28] H. Yang, E. Wang, W. Dong, *et al.*, Ultrafast fragmentation dynamics of triply charged carbon dioxide: Vibrational-mode-dependent molecular bond breakage, *Phys. Rev. A* **97**, 052703 (2018).
- [29] S. S. Iyengar, H. B. Schlegel, J. M. Millam, *et al.*, *ab initio* molecular dynamics: Propagating the density matrix with gaussian orbitals. ii. generalizations based on mass-weighting, idempotency, energy conservation and choice of initial conditions, *J. Chem. Phys.* **115**, 10291 (2001).
- [30] H. B. Schlegel, J. M. Millam, S. S. Iyengar, *et al.*, *ab initio* molecular dynamics: Propagating the density matrix with gaussian orbitals, *J. Chem. Phys.* **114**, 9758 (2001).
- [31] H. B. Schlegel, S. S. Iyengar, X. Li, *et al.*, *ab initio* molecular dynamics: Propagating the density matrix with gaussian orbitals. iii. comparison with born–oppenheimer dynamics, *J. Chem. Phys.* **117**, 8694 (2002).
- [32] S. Mai, D. Avagliano, M. Heindl, *et al.*, **SHARC3.0: Surface hopping including arbitrary couplings - program package for non-adiabatic dynamics**, <https://sharc-md.org/> (2023).
- [33] S. Mai, P. Marquetand, and L. Cález, Nonadiabatic dynamics: The sharc approach, *WIREs Comput. Mol. Sci.* **8**, e1370 (2018).
- [34] G. Li Manni, I. Fdez. Galván, A. Alavi, *et al.*, The openmolcas web: A community-driven approach to advancing computational chemistry, *J. Chem. Theory Comput.* **19**, 6933 (2023).
- [35] B. O. Roos, P. R. Taylor, and P. E. M. Sigbahn, A complete active space SCF method (CASSCF) using a density matrix formulated super-CI approach, *Chem. Phys.* **48**, 157 (1980).
- [36] B. O. Roos, The complete active space self-consistent field method and its applications in electronic structure calculations, *Adv. Chem. Phys.* **69**, 399 (2007).
- [37] See supplemental material at [url will be inserted by publisher], which includes refs. 43–52.
- [38] R. H. Dalitz, Cxii. on the analysis of τ -meson data and the nature of the τ -meson, *Philos. Mag.* **44**, 1068 (1953).
- [39] M. Poline, A. Dochain, Rosén, *et al.*, Vibrationally-dependent molecular dynamics in mutual neutralisation reactions of molecular oxygen ions, *Nat. Comm.* **16**, 8528 (2025).
- [40] A. Bogot, M. Poline, M. Ji, *et al.*, Unravelling non-adiabatic pathways in the mutual neutralization of hydronium and hydroxide, *Nat. Chem.* **17**, 541 (2025).
- [41] M. Poline, A. Dochain, S. Rosén, *et al.*, Mutual neutralization of NO^+ with O^+ , *Phys. Rev. Lett.* **132**, 023001 (2024).
- [42] D. Liu, C. Zhang, *et al.*, Data for: On-the-fly non-adiabatic molecular dynamics reveals dissociation mechanisms of multiply-charged molecules, [10.5281/zenodo.18831135](https://zenodo.org/record/18831135) (2026).
- [43] J. C. Tully, Molecular dynamics with electronic transitions, *J. Chem. Phys.* **93**, 1061 (1990).
- [44] J. E. Subotnik, A. Jain, B. Landry, A. Petit, W. Ouyang, and N. Bellonzi, Understanding the surface hopping view of electronic transitions and decoherence, *Annu. Rev. Phys. Chem.* **67**, 387 (2016).
- [45] D. E. Woon and T. H. Dunning, Gaussian basis sets for use in correlated molecular calculations. V. core-valence basis sets for boron through neon, *J. Chem. Phys.* **103**, 4572 (1995).
- [46] G. Granucci, M. Persico, and A. Zocante, Including quantum decoherence in surface hopping, *J. Chem. Phys.* **133**, 134111 (2010).
- [47] B. Siegmann, U. Werner, H. O. Lutz, *et al.*, Complete coulomb fragmentation of CO_2 in collisions with $5.9 \text{ meV u}^{-1} \text{ Xe}^{18+}$ and Xe^{43+} , *J. Phys. B* **35**, 3755 (2002).
- [48] M. Rosenblatt, Remarks on some nonparametric estimates of a density function, *Ann. Math. Statist.* **27**, 832 (1956).
- [49] J. Ullrich, R. Moshhammer, A. Dorn, *et al.*, Recoil-ion and electron momentum spectroscopy: reaction-microscopes, *Rep. Prog. Phys.* **66**, 1463 (2003).
- [50] X. Ren, J. Zhou, E. Wang, *et al.*, Ultrafast energy transfer between π -stacked aromatic rings upon inner-valence ionization, *Nat. Chem.* **14**, 232 (2022).
- [51] S. Jia, J. Zhou, X. Wang, *et al.*, Cold-target electron-ion-coincidence momentum-spectroscopy study of electron-impact single and double ionization of N_2 and O_2 molecules, *Phys. Rev. A* **107**, 032819 (2023).
- [52] J. Zhou, X. Yu, S. Luo, *et al.*, Triple ionization and fragmentation of benzene trimers following ultrafast intermolecular coulombic decay, *Nat. Comm.* **13**, 5335 (2022).